

Poisson — a field on the unit square

A manufactured exact solution, homogeneous Dirichlet data and a five-point discretisation.

01 / MODEL + CONDITIONS

$$-\Delta u = f$$
$$u = 0$$
$$f(x,y) = 2\pi^2 \sin(\pi x) \sin(\pi y),$$

Find $u \in H_0^1(\Omega)$:

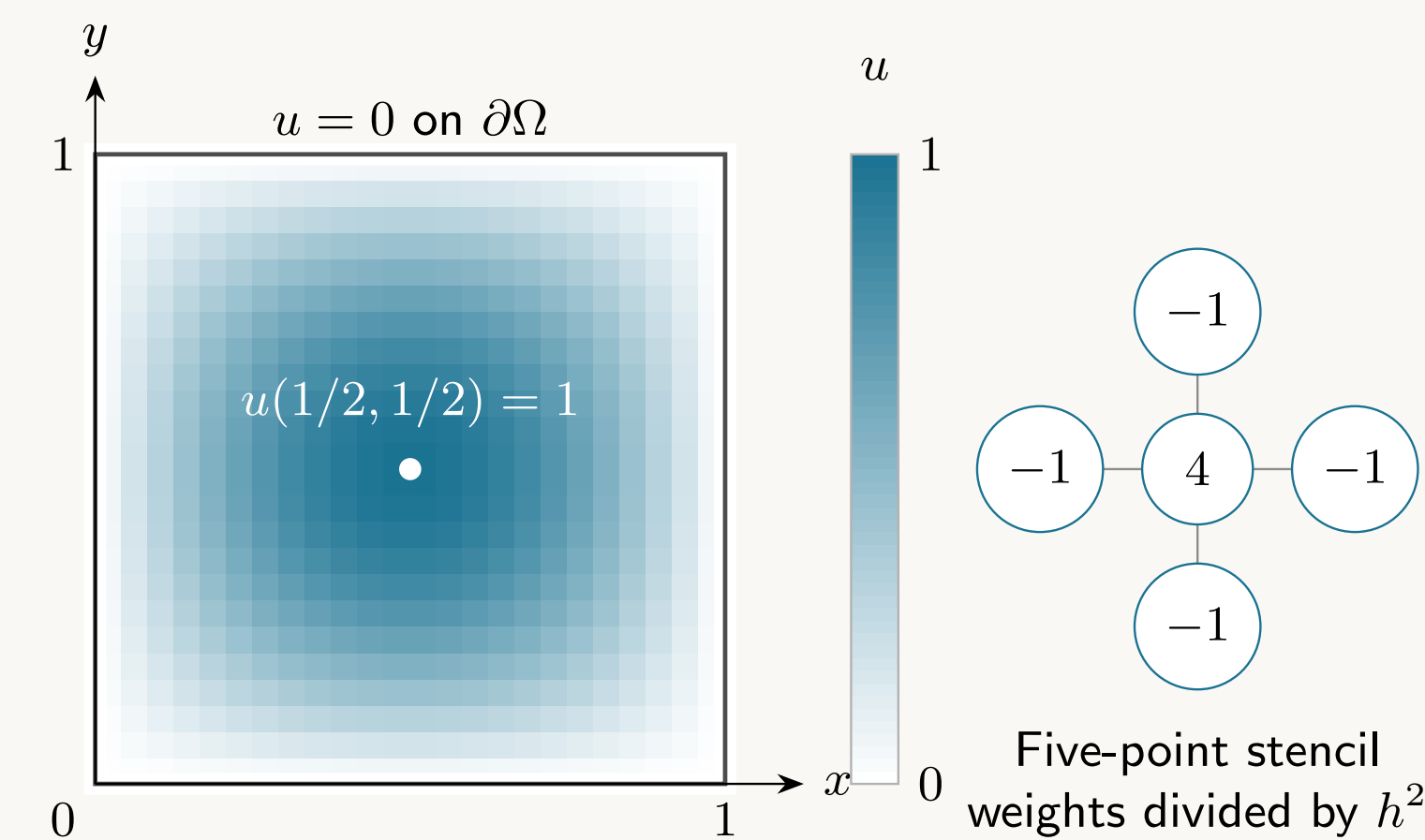
$$\int_{\Omega} \nabla u \cdot \nabla v \, d\Omega = \int_{\Omega} f v \, d\Omega \qquad \forall v \in H_0^1(\Omega).$$

$$\text{in } \Omega = (0,1)^2,$$
$$\text{on } \partial\Omega,$$

02 / EXACT SOLUTION + CHECK

$$u(x,y) = \sin(\pi x) \sin(\pi y),$$
$$u_{xx} = -\pi^2 u, \qquad u_{yy} = -\pi^2 u,$$
$$-\Delta u = 2\pi^2 u = f, \qquad u|_{\partial\Omega} = 0,$$
$$u(\tfrac{1}{2}, \tfrac{1}{2}) = 1, \qquad \max_{\bar{\Omega}} u = 1,$$
$$\int_{\Omega} |\nabla u|^2 \, d\Omega = \int_{\Omega} f u \, d\Omega = \frac{\pi^2}{2}.$$

03 / GEOMETRY + BEHAVIOUR



Exact solution sampled at 24×24 cell centres; no numerical solve.

04 / DISCRETISATION

$$x_i = ih, \ y_j = jh, \ h = 1/N,$$
$$\frac{4u_{ij} - u_{i-1,j} - u_{i+1,j} - u_{i,j-1} - u_{i,j+1}}{h^2} = f_{ij},$$
$$1 \leq i,j \leq N-1,$$
$$A\mathbf{u} = \mathbf{f}, \quad A = A^T, \quad \mathbf{z}^T A \mathbf{z} > 0 \ (\mathbf{z} \neq 0),$$
$$\mathbf{r} = \mathbf{f} - A\mathbf{u}_h,$$
$$\|u_h - u\|_{\infty} = O(h^2) \quad \text{for this smooth solution.}$$

Distinguish the algebraic residual from discretisation error; check boundary values and grid convergence.

Select and double-click any equation or diagram to edit its source. Scroll to work below.